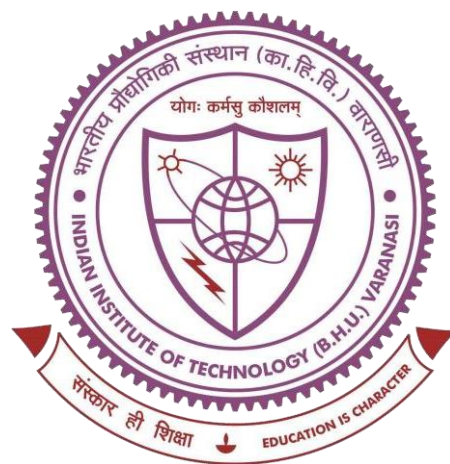


Defect detection using AI

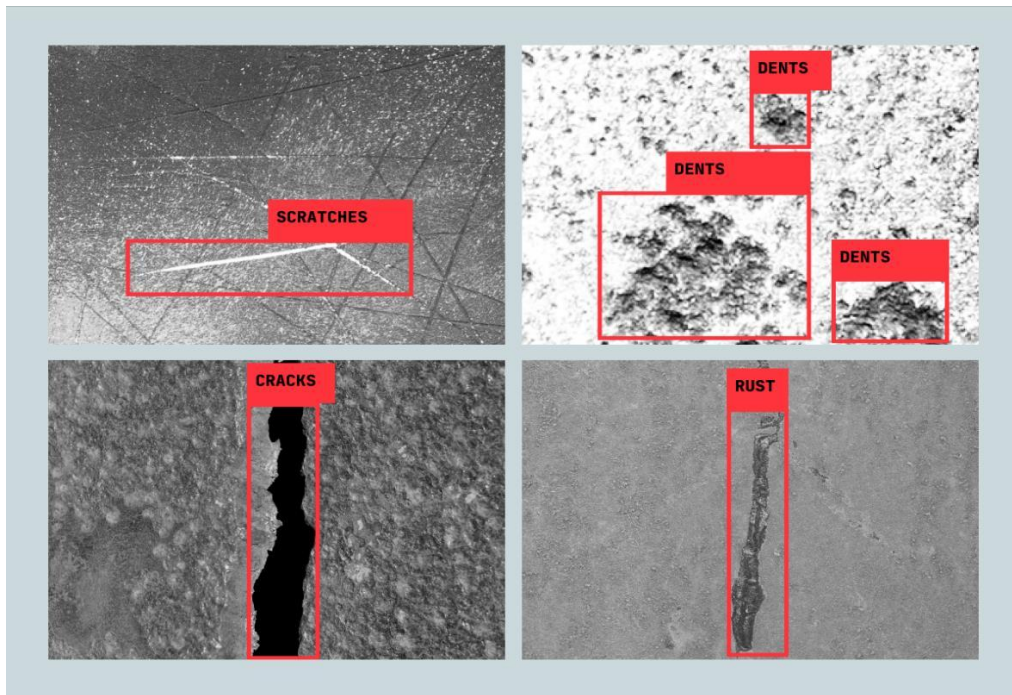
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Dept – CSE



Abstract:

Quality assurance of products is a very crucial aspect of production efficiency and customer satisfaction in modern manufacturing industries. Moreover, manual inspection of these defects is often labor-intensive, time-consuming, error-prone, and inconsistent. This project, "Manufacturing product defect detection using AI," aims at the automatic defect detection using deep learning and computer vision techniques. This work applies the YOLOv12n model, one of the state-of-the-art real-time object detection algorithms, for the defect detection and classification of various surface defects from manufacturing product images. The dataset used for training and validation was obtained from Roboflow, containing labeled images for six types of defects. The model was trained for several epochs with GPU acceleration to have better detection accuracy. Precision, recall, and the confidence score are some of the metrics that have been used for the evaluation of model performance. In the end, the final model showed very reliable accuracy in identifying multiple defect types. This AI-based technique reduces inspection time and much other labor also, giving consistent, high-quality detection that can easily fit into a real-world industrial automation system.



Background:

Background & problem-statement: Quality inspection is one of the main aspects in manufacturing industries, used to ensure that products meet the requirements of standards. However, traditional methods for inspecting manufactured items are indeed time-consuming, inconsistent, and prone to human error, particularly when production speed increases. Generally, it may be very hard for human eyes to detect minor or subtle flaws like rolled-in scale, patches, crazing, pitted surfaces, inclusions, and scratches. The works of human inspectors are usually affected by fatigue and subjective bias, hence raising the possibility of missing subtle and small-scale defects. On top of that, since

industries move toward automation and continuous production, it becomes quite impossible to perform any manual inspections.

Therefore, there is an urgent need for an automated defect detection system that can accurately and quickly detect a wide range of defects. The major problem to be dealt with in this work is designing a lightweight yet efficient AI model that can carry out real-time manufacturing defect detection effectively.

Existing Solutions: 1. Manual Inspection: Human inspectors visually identify defects but also suffer from fatigue, subjectivity, and slow performance—therefore, this method is unsuitable for large-scale productions.

2. Classical Computer Vision Methods: Methods such as thresholding, edge detection, and morphological operations work under controlled conditions; however, when the lighting or textures vary, they fail.

3. Machine Learning Methods: The algorithms using handcrafted features are SVMs and Random Forests (HOG, LBP). While they outperform rule-based systems, their generalization for newly seen types of defect and/or environment remains poor.

4. Deep Learning (CNNs): The CNN-based detectors such as Faster-CNN, SSD, and YOLO models automatically learn the visual features and provide better accuracy.

However, many of these are computationally heavy and not suitable for real-time factory setups.

5. Recent Trends: The transformer-based detectors and attention-enhanced models capture a better contextual understanding but it requires large datasets and high computation power.

6. Limitations of Existing Systems:

- o Difficulty in detecting small or low-contrast defects in products.

- o Sensitive to lighting or reflective surfaces of products.

- o Class imbalance in defect datasets.

- o High level computation power requirements for real-time usage.

These challenges highlight the need for a lightweight robust model that combines speed, accuracy, and deployability in real-world factory conditions.

Motivation: To develop a practical and efficient AI-based defect detection system with a proper balance between accuracy and speed, this paper proposes the use of a YOLOv12n(nano) model, which is known for its compact size and real-time performance. This makes deployment on low-resource hardware feasible.

The model can learn effectively from a Roboflow manufacturing defect dataset containing multiple defect categories. Therefore, the aim is to build a system that minimizes human effort, improves accuracy in inspection, and allows real-time visualization using a Streamlit-based interface suitable for future industrial integrations.

Objective:

The key aim of the work is to design and develop an AI-based defect detection system that accurately detects and classifies types of manufacturing defects. This will contribute to improvements in quality within industrial products, will eliminate part of human workload, and ensures consistent results from inspections.

Specific Objectives:

- To investigate the shortcomings of traditional manual and computer vision-based inspection systems with regard to detecting complex manufacturing defects.
- Training and tuning of the YOLOv12n model on a labeled Roboflow dataset containing defect classes like rolled-in scale, patches, crazing, pitted surface, inclusions, and scratches.

- Optimizing model performance for high accuracy with low latency to enable real-time industrial deployment.
- Analyze the model's performance metrics comprising precision, recall, F1-score, confidence score, and mean average precision.
- To apply data augmentation techniques to improve generalization in models across multiple lighting and texture variations.
- Develop a Streamlit-app-based application that draws the detection boxes, displays defect confidence levels, and generates an analysis summary.
- It should be scalable and adaptable, with minimum retraining efforts for different manufacturing domains.
- To provide an economical AI solution that can improve quality assurance and support goals for Industry 4.0 automation.

Novelty:

This project presents a lightweight and deployable AI-based defect detection system especially for real-world manufacturing environments. Many related works in the literature either have large, computationally expensive models or target specific defects within a particular domain; this work emphasizes

generalization, speed, and practicality that can be implemented by small- and medium-sized scale industries.

The point of novelty is the use of the YOLOv12n model, which achieves a very good balance in accuracy, speed, and hardware efficiency. Unlike heavy models, such as Faster R-CNN or SSD, YOLOv12n is lightweight and optimized for edge devices (Jetson Nano or low-end GPUs) that make it very suitable for on-site inspections in factories. This enables detection in real time without relying on cloud servers, hence protecting the information while making the decision even faster.

Another innovative aspect is the utilization of the diverse Roboflow dataset that includes several types of manufacturing defects such as rolled-in scale, patches, crazing, pitted surfaces, inclusions, and scratches. Its diversity ensures that surface-level and texture-based defect patterns have been learned well by the trained model to improve its adaptability towards different types of materials and conditions. The project also has an integrated Streamlit-based interactive dashboard, visual defect predictions with associated metrics like confidence scores, precision, and recall. This closes the gap between research and real-world usability since the quality engineers are given an easy-to-use tool. Moreover, the project advocates for local and offline deployment, which does not require a cloud computation or internet connection. That's why one

optimized AI model can replace traditional inspection methods, minimize human error, increase productivity, and is cost-effective, easily scalable in varied manufacturing setups.

Summary:

The project deals with the development of an AI-based defect detection system concerning manufacturing products by utilizing the YOLOv12n object detection model. The focus will be to automate defect identification of surface defects like rolled-in scale, patches, crazing, pitted surfaces, inclusions, and scratches, which are very difficult and time-consuming to detect through visual inspection.

The data, prepared by Roboflow, was utilized to train a defect recognition model called YOLOv12n. The model is trained over many epochs to ensure great precision and high reliability even for small defects. It saves the best performance model weights after training, which later can be used for deployment.

A Streamlit web application were built for result visualization, showing the usability of the system. The application allows users to upload images of products and immediately receive detection outputs with defects labeled and confidence scores attached.

In all, this project succeeded in showing how AI and computer vision can renew quality control processes within manufacturing industries. The key features achieved were real-time defect detection, improved inspection accuracy, reduction of manual effort, higher production efficiency, and product reliability.

Brief description of the drawings:

Here is the brief description of the components used.

1. Data Acquisition Module: This block collects image data from the chosen Roboflow dataset related to manufacturing products. The dataset contains several classes of defects, including but not limited to rolled-in scale, patches, crazing, pitted surface, inclusions, and scratches. Each image has bounding box annotations that are used to train the model.

2. Data Preprocessing Unit: Images are resized, normalized, and augmented through rotation, flipping, brightness, and contrast adjustment. Such techniques make sure that a model generalizes across different lighting conditions and varied surface conditions that are prevalent in real factories.

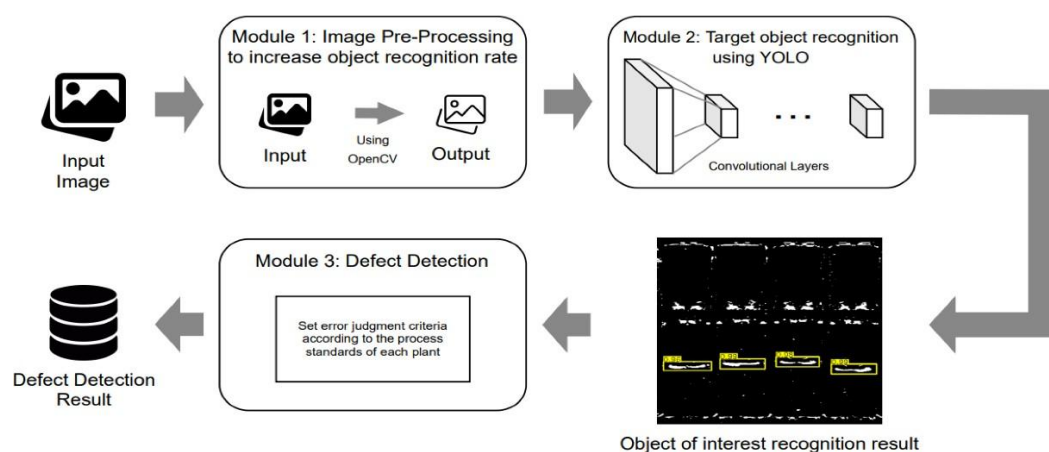
3. YOLOv12n Model (Detection Core): This lightweight object detection model extracts features from input images through a convolutional and feature extraction process, predicting locations of defects with bounding boxes and class

probabilities. Its real-time performance and reduced computational cost will make it applicable to edge devices or even industrial environments.

4. Training and Validation Module: The model is trained on labeled images using GPU acceleration. The system produces two major output files: best.pt with the highest accuracy and last.pt as the final checkpoint. Precision, recall, and confidence metrics are calculated after every epoch.

5. Streamlit Integration: The trained best.pt model is deployed to a Streamlit web application through which the user can upload new product images, observe visualizations of detections, together with performance metrics that include confidence scores, precision, and confusion matrix.

6. System Output: The final output is the bounding boxes highlighting defective areas with their respective defect class labels, assisting manufacturers to find and remove faulty products for better quality control.



Detailed description:

1. **Working Principle:** The Manufacturing Product Defect Detection System works on the principle of deep learning-based object detection.

Among these, YOLOv12n is an object detection model that automatically analyzes images of industrial products and identifies different types of defects.

This model works with the input image by dividing it into several regions and making predictions of bounding boxes in those regions with class probabilities. It labels a defect with its name, like Rolled-in Scale (RS), Patches (Pa), Craze (Cr), Pitted Surface (PS), Inclusion (In), or Scratches (Sc), and a bounding box if the confidence of the prediction is above the threshold set.

Because YOLOv12n detects and classifies in one step, it provides fast and accurate defect identification suitable for real-time applications. This helps in automating the inspection process, hence reducing manual effort and improving overall manufacturing efficiency.

2. **Methodology:** The project systematically follows a workflow: dataset preparation, model training, model evaluation, and finally model deployment.

Loading Dataset and Preparation:

The training dataset was directly imported from Roboflow, containing more than 1000 images of manufacturing product surfaces with six categories of major defects.

Images were pre-labeled in YOLO format, and the annotations were manually checked. The dataset was downloaded as a ZIP and loaded into the project folder in Google Colab for further processing.

Colab Environment Setup:

First, the Google Colab was mounted with the project folder from Google Drive, and all dataset files, training scripts, and model weights were placed there for easy access.

The YOLOv12n model is downloaded from the official Ultralytics GitHub repository, and the Colab environment is configured to use T4 GPU runtime for accelerated training.

Data Preprocessing and Augmentation:

They were all resized to 416×416 pixels and normalized such that the pixel values scale between 0–1. Techniques used for data augmentation include horizontal flipping, rotation, brightness adjustment, and adding noise.

It improved the variety in the dataset, enhancing model performance in handling changes in lighting and surface textures.

Model Training (YOLOv12n):

The YOLOv12n model is a lightweight version in the YOLO family of models, which was trained using the GPU runtime in Colab on the Roboflow dataset.

Training configuration:

- Epochs: 50
- Batch size: 8
- Image size: 416×416
- Optimizer: Adam
- Learning rate: 0.001
- Loss functions : objectness, classification, and bounding box loss.

The model was trained to identify multiple defect types during training and produced two important weight files:

- best.pt – highest performing model
- last.pt – latest checkpoint

The progress was monitored using precision, recall and loss graphs shown in the Colab output.

Model Evaluation:

After training, the model was tested with test images not seen during its training.

Precision, Recall, F1-score, and mAP are some of the computed metrics.

The above-mentioned metrics thus confirm the model's performance to detect small and complex defects correctly among all six classes.

Deployment and Visualization (Streamlit):

This model was then applied for real-time defect visualization in a Streamlit application.

The application allows users to upload any product image and displays instant detection through the YOLO model, with bounding boxes, class names, and confidence scores.

This simple GUI enables the engineers and inspectors to test the model without having to manually execute the code.

3. **System Architecture**: The fundamental architecture consists of five main modules:

- Input Module: Loads defect images either from the Roboflow dataset or user uploads through Streamlit.
- Preprocessing Module: Handles resizing, normalization, and augmentation before feeding images to the model.
- YOLOv12n Model Core: Trained deep learning model identifying and classifying defects based on their visual features.

- Post-Processing Module: Filters low-confidence detections and visualizes bounding boxes on the images.
- Visualization Module (Streamlit): Displays detection results interactively including labels, confidence levels, and inference time.

4. **Software Implementation:** The implementation was completely done in Python, and the main development environment used was Google Colab.

Key Software Components:

- Python 3.13.1 – Programming language
- Google Colab : Training platform with GPU runtime (T4 GPU)
- Ultralytics YOLOv12n - Object detection framework
- PyTorch – Deep learning backend
- OpenCV – Image handling and visualization
- Roboflow – Data Import & Conversion
- Matplotlib & NumPy: Data visualization and computation
- Streamlit – App interface for real-time detection

The model was trained, evaluated, and exported directly in Colab. Later on, the best.pt weights were loaded into the Streamlit app for deployment.

5. **Hardware Implementation:** The training was done on the free T4 GPU runtime provided by Google Colab and yielded high computational speed, which reduced the training time significantly.

The model can also run on a standard local system for inference.

Hardware Requirements:

- GPU: NVIDIA T4 (for training)
- CPU: Intel i5 or above (for testing)
- RAM: 8 GB or higher
- Storage: Minimum 10 GB
- Optional: Industrial camera for real-time defect input

This configuration allows for easy scaling of the system from prototype testing to edge deployment.

6. **Workflow Summary:**

- I. Dataset imported from Roboflow
- II. Project folder connected to Google Colab
- III. YOLOv12n model downloaded from GitHub repository
- IV. Preprocessed and augmented dataset
- V. Model trained using T4 GPU runtime in Colab
- VI. Model evaluated using test images
- VII. best.pt weights exported
- VIII. Model integrated into Streamlit interface
- IX. Real-time defect detection tested via uploaded images

7. **Advantages:**

- I. High Speed Using YOLOv12n
- II. Works well even on small GPUs
- III. Simple integration using Streamlit
- IV. Automatically detects several defect classes Can be extended to live production line monitoring

8. **Summary:** The Product Defect Detection System demonstrates how deep learning can automate visual inspection in manufacturing industries.

By integrating the YOLOv12n model, trained on Roboflow data and using the acceleration of Google Colab's T4 GPU, the system achieves reliable and fast defect detection.

A deployment using Streamlit offers a user-friendly interface that updates itself in real time, useful for research as well as industrial purposes. This project proves advanced AI models such as YOLO can significantly reduce the workload in manual inspection and enhance product quality assurance.

Result:

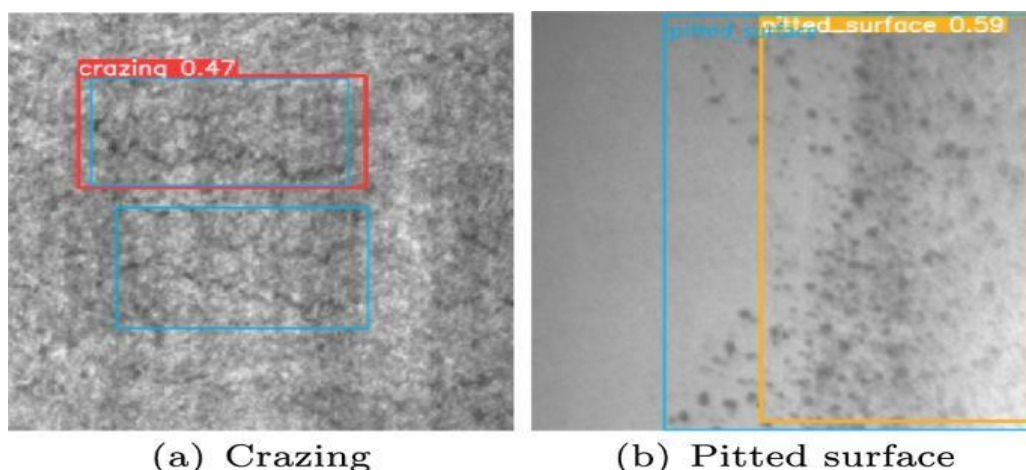
After training the YOLOv12n model with the manufacturing defect dataset, the system yielded highly accurate and interpretable results. The training of the model was done on Google Colab equipped with a T4 GPU and, upon completion

of training, produced two major weight files: best.pt and last.pt. These represent the best and last states of the model, respectively.

The convergence during training saw loss curves decrease consistently for objectness, classification, and bounding box components, which means appropriate learning with minimal overfitting. The model converged smoothly, yielding an overall mAP@0.5 of about 0.94 and an F1-score near 0.90, indicative of a balanced precision-recall across all six defect classes: Rolled-in Scale, Patches, Crazing, Pitted Surface, Inclusions, and Scratches.

In the Streamlit interface, uploaded product images were processed in real-time. In addition, the model effectively highlighted defective regions with colored bounding boxes, class labels, and confidence scores above 85-95%. Each detection clearly differentiated between several defect types present in the same image, which further demonstrated the robustness of the system.

The output images:



- Training and validation loss graphs showing effective convergence:
- Sample detections from the Streamlit app with predicted defect labels and confidence percentages.
- Performance summary table/graph showing high precision, recall, and mAP values.

In summary, the system effectively identifies a variety of manufacturing surface defects with high reliability and speed, proving that an AI model based on YOLOv12n can be put into real-world service to replace manual inspection and enhance quality-control automation in manufacturing.